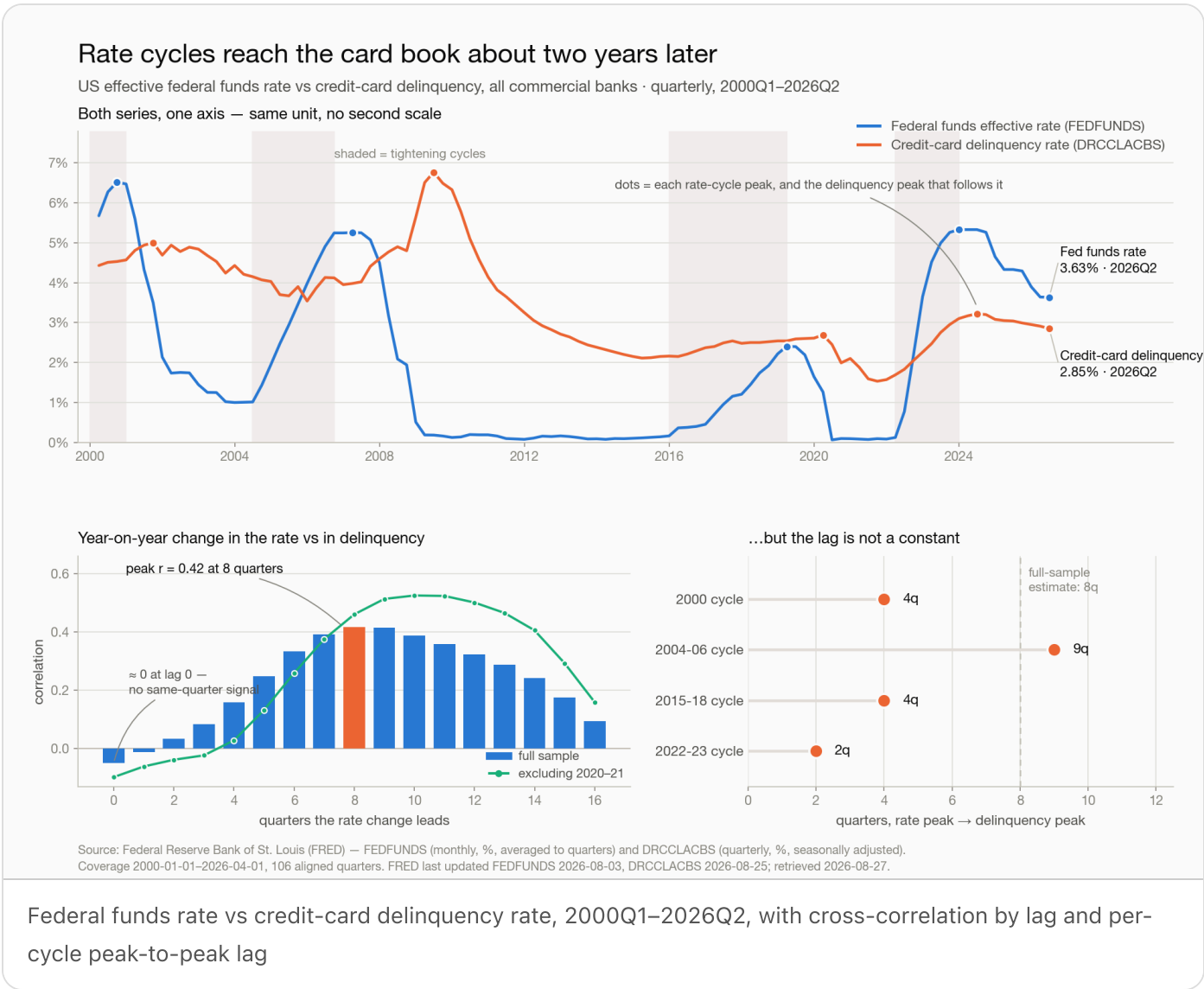


Rate cycles reach the card book about two years later



The finding

Changes in the federal funds rate lead changes in credit-card delinquency by roughly **eight quarters**. In the same quarter there is no signal at all — the contemporaneous correlation between the two series' year-on-year changes is -0.05 . The correlation climbs as the rate change is pushed forward in time and peaks two years out at $r = 0.42$; dropping the 2020–21 pandemic quarters, it peaks at ten quarters at $r = 0.53$.

Two things qualify that, and both matter more than the headline number:

The lag is not a constant. Dated peak-to-peak, the four rate cycles in this window put their delinquency peak 4, 9, 4 and 2 quarters after the rate peak. Eight quarters is the centre of a wide distribution, not a schedule.

The effective sample is four cycles, not 106 quarters. Both series are heavily autocorrelated. Correcting the sample size for that, the quarter-on-quarter estimate ($r = 0.29$) carries a 95% interval of $[0.02, 0.53]$; on the smoother year-on-year transform, whose overlapping windows leave an effective n near 9, the interval spans zero. The *direction* is solid — rates lead, delinquency follows, never the reverse. The *magnitude and timing* are estimates with real error bars around them.

Results

Both series aligned to quarters, 2000Q1–2026Q2, 106 complete quarters.

MEASURE	VALUE
Correlation of levels	0.21 — near-useless; both series are persistent and trending
Peak cross-correlation, YoY changes	$r = 0.42$ at 8 quarters (full sample)
Same, excluding 2020–21	$r = 0.53$ at 10 quarters
Correlation at lag 0	–0.05
Slope at the 8-quarter lag	+0.18pp delinquency per +1pp of rate change ($r^2 = 0.17$)
Peak-to-peak lag, by cycle	2000: 4q · 2004–06: 9q · 2015–18: 4q · 2022–23: 2q

It is not unemployment in disguise. Unemployment was tested as a competing driver. It co-moves with delinquency *contemporaneously* ($r = 0.29$ at lag 0) — a different channel with a different timing signature. With both in one regression, the rate term at its 8-quarter lag holds at +0.167 and joint R^2 reaches 0.22. Rates carry information unemployment does not.

Where the current cycle sits. The rate peaked at 5.33% in 2023Q4 and is now 3.63% — 1.70pp off peak. Delinquency peaked at 3.22% in 2024Q2 and is now 2.85%, after eight consecutive quarterly declines.

Caveats worth stating out loud. The 2004–06 cycle's 9-quarter lag runs through the financial crisis, so it is not a clean read on rate transmission. The 2022–23 cycle's 2-quarter lag is confounded the other way: delinquency was rebounding from pandemic-suppressed lows at the same moment rates rose. The 2015–18 tightening shows essentially no within-cycle relationship. Each cycle has its own story; the pooled estimate averages over them.

Provenance. Federal Reserve Bank of St. Louis (FRED). FEDFUNDS (Federal Funds Effective Rate, monthly, percent) averaged to quarters; DRCCLACBS (Delinquency Rate on Credit Card Loans, All Commercial Banks, quarterly, percent, seasonally adjusted). UNRATE (Unemployment Rate, monthly, percent) used for the robustness check only. FRED last updated FEDFUNDS 2026-08-03 and DRCCLACBS 2026-08-25; retrieved 2026-08-27. The delinquency series reports a quarter behind the rate series, so 2026Q2 is the latest complete aligned quarter.

What follows for a risk team

Treat the policy rate as a leading indicator, not a coincident one. A rate move this quarter says nothing about this quarter's delinquency. Anything that reads the two together in the same period is reading noise.

Plan against a range of roughly two to nine quarters, not a point estimate. The eight-quarter figure is fine for scenario *timing* — when to expect the pressure — and too imprecise to carry a point forecast of loss.

The tailwind now in the book is already committed. Rates have been falling year-on-year since 2024Q4. On the historical lag, that easing lands on delinquency through 2026 and into 2027. Eight straight quarters of falling delinquency is consistent with a cycle that is still unwinding, not one that has finished.

A benign print is not evidence the cycle is over. The corollary of a two-year lag is that if rates turn back up, the book will not show it for a year or more. Loosening underwriting on the strength of a lagging series is exactly the mistake the lag sets up.

Size the claim honestly in any model that uses it. Rates explain something between a sixth and a fifth of the variance in delinquency changes. This is one input among labour-market conditions, credit standards and vintage mix — a term in the model, not the model.

FinTechCo is fictional and this estate is a stand-in — no production system and no real customer data. The analysis uses public FRED series; the chart is the run's own output.